

ARM-EBM equivalence

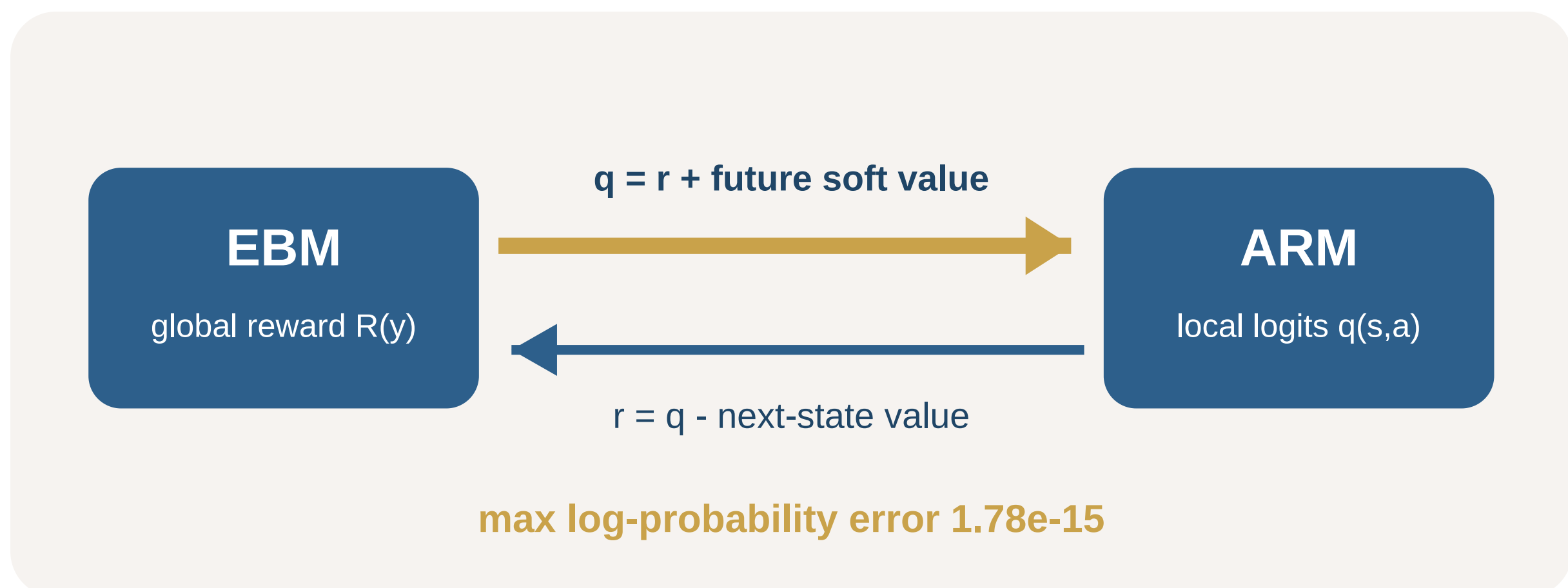
reproduces exactly

Independent CPU reproduction of Blondel et al. (ICML 2026, arXiv:2512.15605)

Exact tabular checks · $V=8$ · $T=4$ · 585 EOS responses + 4,096 fixed-length sequences

Outcome

The paper's core finite-state result holds in an exhaustive CPU implementation. Mapping arbitrary variable-length EBM rewards backwards into EOS-aware ARM logits reproduces every response probability to **1.78e-15**; the ARM → reward → ARM fixed-point residual is **6.66e-16**.



The exact backward map is the acyclic soft Bellman equation; its inverse removes the next-state log-partition.

What was tested

- **Claim 1.** Exact EBM → ARM probability preservation on all 585 valid variable-length EOS responses.
- **Claim 2.** Soft Bellman ARM → reward → ARM round trip over every valid state and action.
- **Claim 3.** Analytical ARM and EBM NLL gaps are exactly zero on both paper sequence spaces.
- **Claim 4.** 80/80 directional KL inequalities pass; worst observed KL/bound ratio: 0.0716.
- **Cost mechanism.** Sequence enumeration grows from 4,096 paths at $T=4$ to 262,144 at $T=6$ for $V=8$.

Interpretation

The ARM's next-token logits encode a *future soft value*. In the fixed-horizon tree, the backwards Bellman recursion supplies that value exactly, so ancestral ARM sampling and globally normalized EBM sampling define the same distribution.

- **Scope.** This is an exact function-space reproduction of the paper's Appendix-C synthetic special case.
- **Not claimed.** It does not reproduce causal/non-causal Transformer training curves or establish LLM-scale behavior.
- **Cost.** The complete reported CPU run, including both NLL optimizations, took 9.1 seconds.

Reproduction bundle

- **Code.** Exact ARM/EBM mapping, optimization runner, and regression tests.
- **Evidence.** Raw NLL and KL-bound CSVs, JSON report, and an HTML convergence plot.
- **Rerun.** `./reproduce_cpu.sh` recreates the environment and all reported outputs.

Machine-readable evidence

- **report.json** — configuration, environment, verdicts, and headline errors for all five claims.
- **training_history.csv** — exact ARM and EBM expected-NLL gaps through 3,000 updates.
- **kl_bound_trials.csv** — both directional KL values, infinity-norm error, bound, and ratio for every trial.

1.78e-15

maximum EBM/ARM log-probability error

80 / 80

directional KL-bound checks passed

~10 s

complete CPU run time